Assignment 4 BS

Group 2
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1 PCA-based Factor Model

In accordance with the paper "Statistical Arbitrage in the U.S. Equities Market" by Avellaneda & Lee (2008), we decompose the returns of each asset using a statistical factor model. The model assumes that the returns $R_{i,t}$ can be explained by a set of common systematic factors and an idiosyncratic residual component:

$$R_{i,t} = \alpha_i + \sum_{j=1}^{K} \beta_{i,j} F_{j,t} + \epsilon_{i,t} \tag{1}$$

where $F_{j,t}$ represents the j-th risk factor, $\beta_{i,j}$ the loadings, and $\epsilon_{i,t}$ the idiosyncratic residual.

1.1 Eigenfactor Construction

To isolate systematic risk, we perform Principal Component Analysis (PCA) on the correlation matrix of asset returns. Following the specific requirements of the assignment, we select the first $K = 4$ principal components. While the paper defines factors through a summation, our implementation utilizes the equivalent matrix representation for computational efficiency. Following the formula (9) of the paper, the matrix of eigenfactors $\mathbf{F}$ is constructed as:

$$\mathbf{F} = \tilde{\mathbf{R}} \mathbf{V}_K \tag{2}$$

where $\tilde{\mathbf{R}}$ is the matrix of standardized returns (calculated as $R_{i,t}/\sigma_i$) and $\mathbf{V}_K$ represents the matrix of the first K eigenvectors. This formulation ensures that the factors are purely statistical representations of common market fluctuations.
To extract the idiosyncratic component $\epsilon_{i,t}$ and the drift term α_i , we perform a rolling OLS regression. For the estimation, we utilize the `np.linalg.lstsq` solver. This choice is motivated by numerical robustness.

1.2 Variance Diagnostics

The statistical validity of the $K = 4$ choice is analyzed through the eigenvalue spectrum and cumulative variance (see Figure 1).

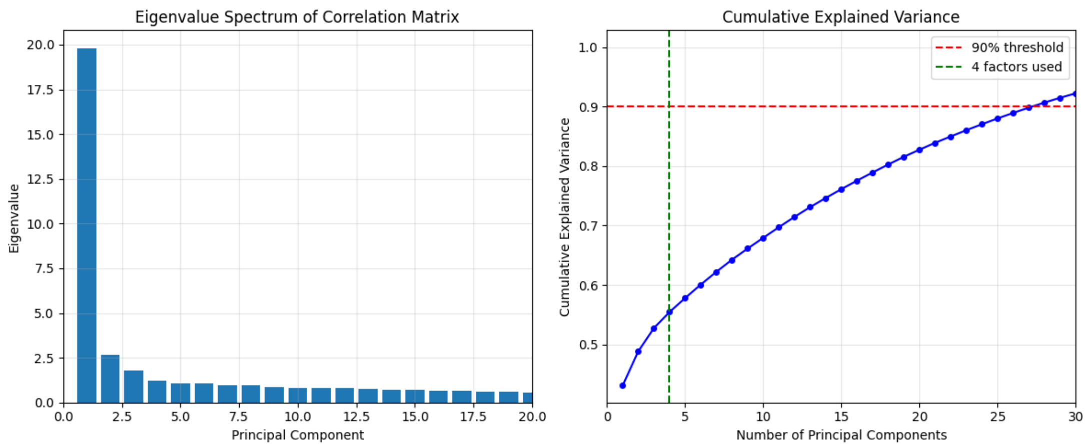

Figure 1: PCA Factor Analysis.

The eigenvalue spectrum shows a dominant first component, representing the market beta, followed by a rapid decay. For this time window, the first 4 factors capture approximately 55% of the total variance, although this fraction is not constant over time.

2 Ornstein-Uhlenbeck Process for Residuals

Following the identification of idiosyncratic residuals $\epsilon_{i,t}$, we model the cumulative residuals $X_t = \sum_{s=1}^{t} \epsilon_{i,s}$ as a mean-reverting Ornstein-Uhlenbeck (O-U) process. This process is defined by the following stochastic differential equation (SDE):

$$dX_t = \kappa(m - X_t)dt + \sigma dW_t \tag{3}$$

To estimate the parameters from discrete market data, we utilize the AR(1) representation of the process:

$$X_{n+1} = a + bX_n + \zeta_{n+1}, \quad n = 1, \ldots, 59 \tag{4}$$

where the discrete coefficients a, b are mapped to the continuous parameters κ, m, σ through the formulas presented in the Appendix of the paper.

2.1 Estimation Methodology

The strategy employs a dual-window approach to balance structural stability with signal responsiveness:

- **PCA Factor Window (252 days):** We use a trailing 252-day window to estimate the correlation matrix and extract the eigenvectors. This longer horizon ensures stable market factors that are not overly sensitive to short-term noise.
- **O-U Estimation Window (60 days):** In accordance with the assignment requirements and the original paper, we use a 60-day window of cumulative residuals to estimate the O-U parameters. This shorter window is designed to capture the local mean-reversion dynamics necessary for generating timely trading signals.

2.2 Parameter Analysis for Selected Assets

The following table summarizes the estimated O-U parameters for the first five assets in our universe, assuming a daily time step ($dt = 1/252$):

Table 1: O-U Parameters for selected assets (60-day window)

Asset	κ	$t_{1/2}$	σ_{eq}	m
ABI.BR	60.15	2.9 days	0.0102	0.0030
AD.AS	15.27	11.4 days	0.0308	0.0304
ADSGn.DE	16.83	10.4 days	0.0193	0.0210
AIR.PA	45.27	3.9 days	0.0239	-0.0129
AIRP.PA	28.12	6.2 days	0.0098	-0.0054

The parameters provide critical insights into the dynamics of the residuals:

- κ**:** Represents the speed of mean reversion. Higher values indicate a stronger "elastic" force pulling the residual back to its mean.
- $t_{1/2}$**:** Calculated as $\ln(2)/\kappa$, it represents the expected time required for the process to dissipate half of its deviation from the mean.
- σ_{eq}**:** The equilibrium volatility of the process, indicating the typical range of fluctuations around the mean.
- m**:** The long-term equilibrium level of the cumulative residuals.

As observed in Figure 2, the cumulative residuals X_t tend to converge toward zero at the end of the estimation window. This behavior is a direct mathematical consequence of the **Ordinary Least Squares (OLS)** construction used in the factor model estimation.

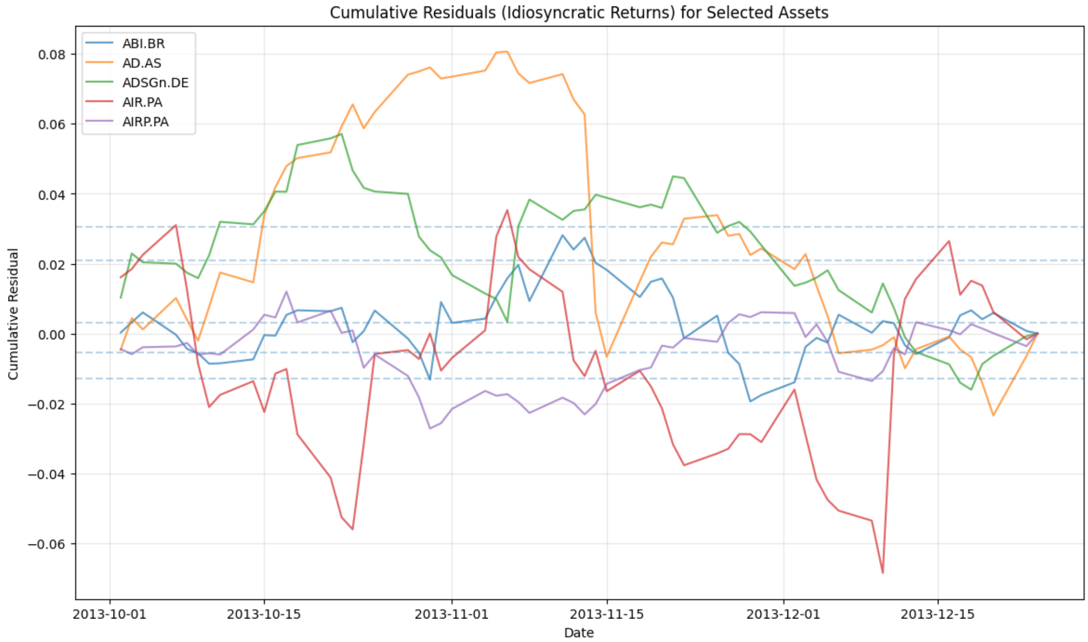

Figure 2: Cumulative Residuals (X_t) for selected assets over the 60-day estimation window

3 S-Score and Trading Signals

To generate actionable trading signals, we filter the asset universe for those exhibiting high mean-reversion speeds, specifically requiring $\kappa > 8.4$. This threshold filters for assets whose mean-reversion time is of the order of 1.5 months. For these valid assets, we calculate the **modified S-score** to determine their relative deviation from the local equilibrium:

$$s_{mod,t} = \frac{X_t - m}{\sigma_{eq}} - \frac{\alpha}{\kappa \sigma_{eq}} \tag{5}$$

where κ, σ_{eq}, and m are the estimated O-U parameters. The term α represents the idiosyncratic drift extracted from the factor regression, which adjusts the score to account for persistent momentum in the residuals, as suggested by the paper.

3.1 Trading Strategy Logic

The strategy is governed by a state machine that manages entries and exits based on specific S-score thresholds. We set the entry barriers at $s_{bo}, s_{so} = 1.25$ and the exit targets at $s_{bc}, s_{sc} = 0.50$. The logic is summarized in the following table:

Current State	Condition	Action / New State
Flat (0)	$s < -1.25$	Open Long (+1)
	$s > 1.25$	Open Short (-1)
	Else	Remain Flat (0)
Long (+1)	$s > -0.50$	Close Position (0)
	Else	Hold Long (+1)
Short (-1)	$s < 0.50$	Close Position (0)
	Else	Hold Short (-1)

Table 2: Logic for Position Management

4 Rolling Backtest of Statistical Arbitrage Strategy

The backtesting framework is designed to simulate a realistic trading environment, ensuring that the performance metrics account for capital constraints, execution delays, and market frictions.

4.1 Gross Exposure and Weighting Scheme

To maintain a consistent risk profile, the portfolio is constructed under a strict **Gross Exposure constraint of 100%**. Mathematically, the constraint is expressed as:

$$\sum_{i=1}^{n} |w_{i,t}| = 1 \tag{6}$$

Given the equal-weighting scheme, the weight for each asset i at time t is calculated as:

$$w_{i,t} = \text{sgn}(s_{i,t}) \cdot \frac{1}{N_{active,t}} \tag{7}$$

where $N_{active,t}$ is the total number of assets satisfying the entry/hold criteria on that specific date.
As illustrated in Figure 3 below, the 100% gross exposure constraint is strictly satisfied throughout the entire backtesting period.

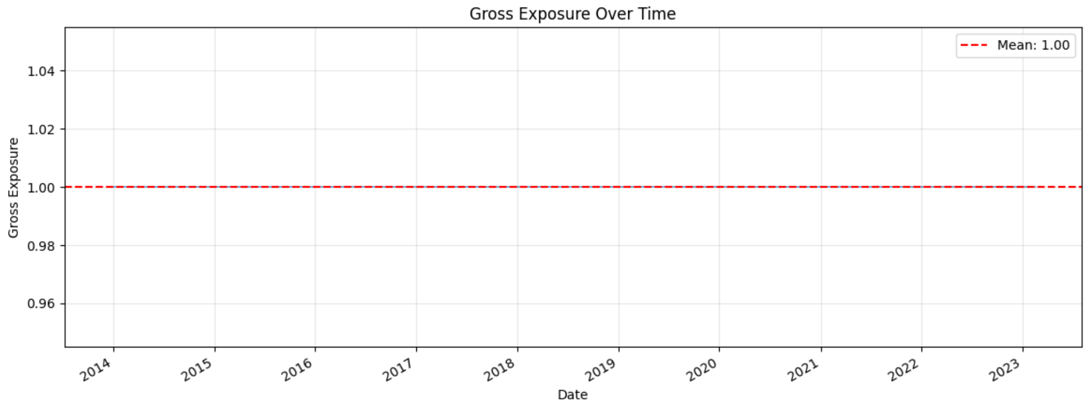

Figure 3: Gross Exposure

4.2 Incorporating Trading Costs

To account for market friction, we implement a transaction cost model based on portfolio **turnover**. This is critical for Statistical Arbitrage due to the frequent rebalancing required as residuals revert to their mean. The total cost for a given rebalancing period is computed by applying a fixed fee rate to the absolute daily change in asset weights:

$$C_t = \text{fee} \times \sum_{i=1}^{n} |w_{i,t} - w_{i,t-1}| \tag{8}$$

4.3 Temporal Alignment and Execution Logic

A fundamental challenge in backtesting is avoiding "look-ahead bias," where information from the future is inadvertently used to make past decisions. We have carefully verified that the backtesting implementation functions exactly as intended, specifically regarding the rigorous temporal synchronization between signal generation and trade execution:

- **Synchronization:** The weights generated by the S-score signals are merged with the return matrix based on common trading dates.
- **One-day Shift:** We apply a forward shift to the weight dataframe. This reflects the operational reality that a signal calculated at the close of day t (using all available data up to that point) can only be executed for the duration of day $t + 1$.

The realized return for the portfolio at time $t + 1$ is thus the product of the weights decided at time t and the returns observed at $t + 1$.

5 Analysis of Strategy Characteristics

This section summarizes the operational and performance metrics of the Statistical Arbitrage strategy compared to a passive Equal Weight benchmark. The results highlight the strategy's ability to generate returns with significantly lower volatility and controlled drawdowns.

5.1 Portfolio Composition and Activity

The strategy was backtested over 2,347 rebalance dates. On average, the model maintains a balanced market-neutral profile, as shown in Table 3. The near-equal distribution of long and short positions suggests that the PCA-based factor model successfully isolated idiosyncratic opportunities while hedging systemic risk.

Metric	Value
Total Rebalance Dates	2347
Average Number of Long Positions	8.9
Average Number of Short Positions	9.1
Average Total Active Positions	18.0

Table 3: Portfolio Position Statistics

5.2 Comparative Performance Analysis

Table 4 presents a side-by-side comparison of the implemented strategy and the Equal Weight benchmark.

Metric	StatArb Strategy	Equal Weight Portfolio
Total Return	49.23%	124.96%
Annualized Return	4.40%	9.10%
Annualized Volatility	6.06%	19.52%
Sharpe Ratio	**0.73**	**0.47**
Maximum Drawdown	-11.59%	-39.93%

Table 4: Performance Comparison: StatArb vs. Equal Weight

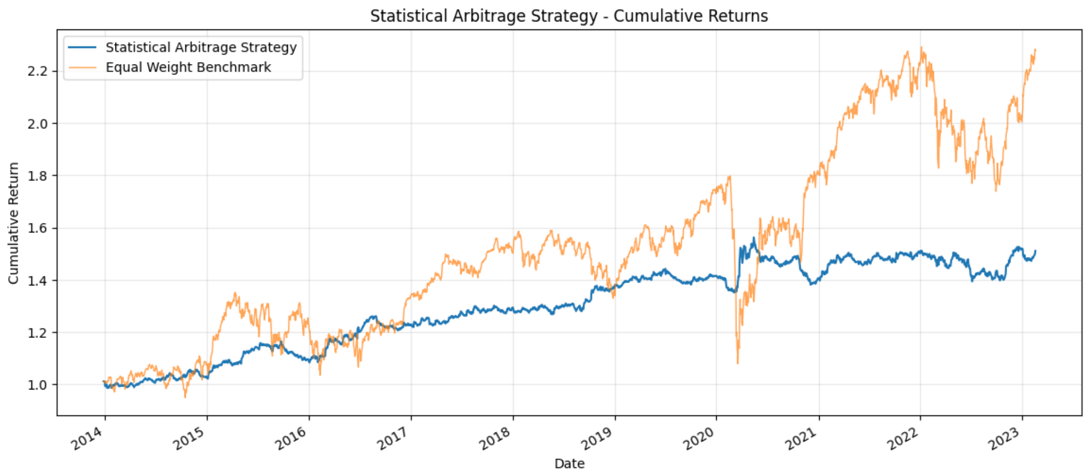

Figure 4: Cumulative returns with different strategies over the 60-day estimation window

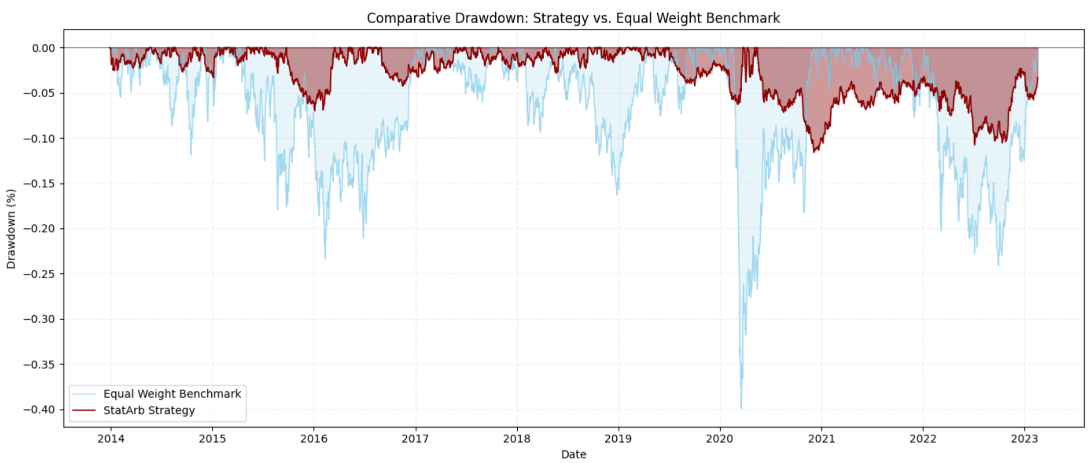

Figure 5: Drawdown comparison with different strategies over the 60-day estimation window

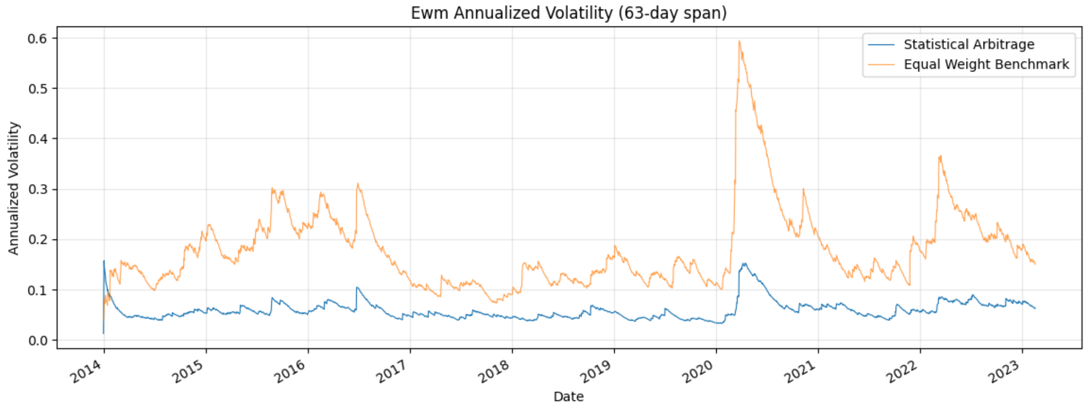

Figure 6: Volatility comparison with different strategies over the 60-day estimation window

5.3 Interpretation of Results

While the Equal Weight portfolio achieved a higher total return, it did so by exposing the capital to significant market beta, resulting in a volatility of 19.52% and a severe maximum drawdown of nearly 40%.
In contrast, the Statistical Arbitrage strategy demonstrates superior risk-adjusted performance, evidenced by a significantly higher Sharpe Ratio (0.73 vs 0.47). The strategy effectively reduced the investment risk; by focusing on mean-reverting residuals rather than market direction, it achieved an annualized volatility of only 6.06% and a contained maximum drawdown of 11.59%.

6 Sensitivity Analysis

6.1 Sensitivity to s-score entry threshold

To evaluate the robustness of the PCA-based statistical arbitrage strategy, we conducted a sensitivity analysis by varying the symmetric opening thresholds s_{bo} and s_{so}. While the closing thresholds were kept constant at $|0.5|$, we tested four different levels for $s_{bo} \in \{1.0, 1.25, 1.5, 2.0\}$ without considering trading costs. The performance metrics for each scenario are summarized in the table below.

Table 5: Strategy Performance Metrics across Different Entry Thresholds

Metric	$s_{bo} = 1.0$	$s_{bo} = 1.25$ (Base)	$s_{bo} = 1.5$	$s_{bo} = 2.0$
Total Return	40.40%	49.23%	40.84%	-32.30%
Annualized Return	3.71%	4.40%	3.75%	-4.11%
Annualized Volatility	5.35%	6.06%	8.13%	13.48%
Sharpe Ratio	**0.69**	**0.73**	**0.46**	**-0.30**
Maximum Drawdown	-14.75%	-11.59%	-22.49%	-43.49%

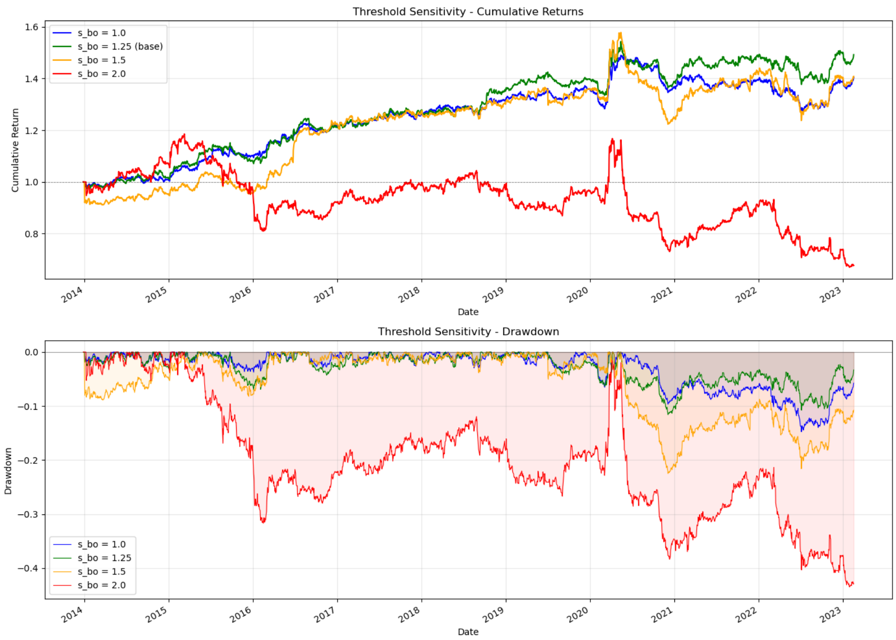

Figure 7: Cumulative Returns and Drawdown with different threshold

The results indicate that the strategy's performance is highly sensitive to the chosen s-score entry level.

The baseline threshold of 1.25 proves to be the most effective, yielding the highest Annualized Return (4.40%) and the best risk-adjusted performance with a Sharpe Ratio of 0.73. It also maintains the lowest Maximum Drawdown (-11.59%), suggesting a superior balance between signal quality and diversification.

Lowering the threshold to 1.0 increases the number of trades (higher turnover) but leads to a decay in efficiency, as shown by the lower Sharpe Ratio (0.69). Statistical arbitrage relies on the assumption that the residual is a mean-reverting Ornstein-Uhlenbeck process. A low threshold triggers trades on small deviations that might just be "noise" rather than a true statistical misalignment.

At $s_{bo} = 2.0$, the strategy fails significantly, resulting in a negative Total Return of -32.30% and a severe Maximum Drawdown of -43.49%. Extreme s-scores often indicate structural breaks (e.g., a company's business model fundamentally changed) or momentum regimes. In these cases, the stock doesn't mean-revert; it continues to trend away, leading to severe losses. Furthermore, the very high annualized volatility (13.48%) at this level indicates a lack of diversification, as very few stocks hit such extreme thresholds simultaneously.

Sensitivity to s-score entry threshold considering trading costs

To further assess the robustness and practical viability of the strategy, we re-executed the sensitivity analysis by introducing all-in trading costs of 5 basis points.

The following table summarizes the key performance metrics across the different entry thresholds $s_{bo} \in \{1.0, 1.25, 1.5, 2.0\}$ after accounting for transaction costs.

Table 6: Strategy Performance Metrics with 5 bps Trading Costs

Metric	$s_{bo} = 1.0$	$s_{bo} = 1.25$ (Base)	$s_{bo} = 1.5$	$s_{bo} = 2.0$
Total Return	2.84%	**10.37%**	4.28%	-50.55%
Annualized Return	0.30%	**1.07%**	0.45%	-7.29%
Annualized Volatility	5.34%	6.05%	8.14%	13.48%
Sharpe Ratio	0.06	**0.18**	0.06	-0.54
Maximum Drawdown	-20.58%	-17.15%	-26.83%	-56.95%

The introduction of trading costs significantly diminishes the performance profile of the strategy.

In particular, the performance of the 1.0 threshold collapses after costs. The higher trade frequency associated with a lower entry barrier generates excessive turnover, causing transaction costs to erode nearly all generated alpha. The Sharpe ratio drops dramatically from 0.69 (frictionless) to 0.06.

The baseline threshold remains the most resilient: it maintains the highest Total Return (10.37%) and the best risk-adjusted profile.

6.2 Fixed factors vs variable factor selection

In this section, we evaluate the impact of using a dynamic number of principal components (PCs) rather than a fixed set of four factors. The number of factors is selected at each rebalancing date to achieve a specific target level of cumulative explained variance: 40%, 55%, 65%, and 75%.

By allowing the number of factors to vary, we aim to capture a consistent proportion of the market's systematic risk over time, regardless of whether the market is in a high-correlation regime (where few factors suffice) or a low-correlation regime (where more factors are needed to explain returns). For each scenario, we re-estimated the residuals, the Ornstein-Uhlenbeck parameters, and the resulting modified s-scores.

To determine the number of factors p for a given variance target η, we analyze the eigenvalues $\{\lambda_j\}_{j=1}^N$ obtained from the PCA. The fraction of total variance explained by the first p components is calculated as the ratio of the sum of the first p eigenvalues to the sum of all eigenvalues:

$$\text{Cumulative Variance}(p) = \frac{\sum_{j=1}^{p} \lambda_j}{\sum_{j=1}^{N} \lambda_j} \tag{9}$$

For each target η, we select the minimum p such that Cumulative Variance$(p) \geq \eta$.

Numerical Results

The performance metrics for the different variance targets (assuming no trading costs) are summarized in the table below:

Table 7: Strategy Performance across Explained Variance Targets

Variance Target	Total Return	Ann. Return	Ann. Volatility	Sharpe Ratio	Max Drawdown
40%	68.34%	5.76%	6.86%	0.84	-15.24%
55%	**74.65%**	**6.18%**	**6.50%**	**0.95**	**-8.93%**
65%	55.57%	4.86%	6.07%	0.80	-10.28%
75%	53.43%	4.71%	5.80%	0.81	-7.71%

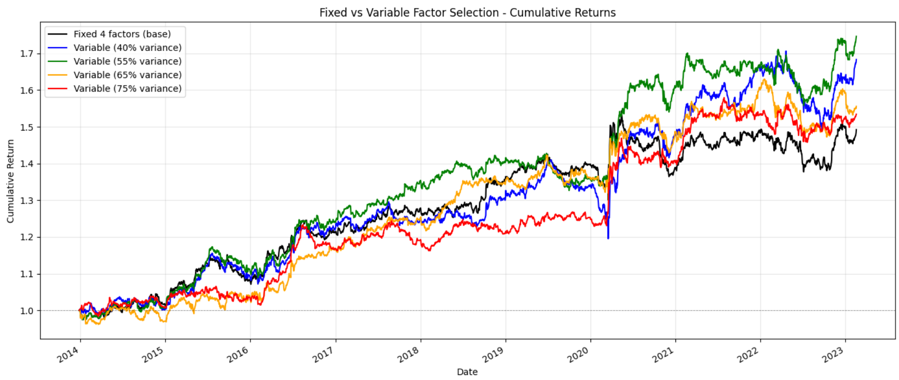

Figure 8: Cumulative Returns with different principal factors.

Discussion and Graphical Analysis

The sensitivity analysis reveals a non-linear relationship between the number of factors and the strategy performance:

- **The Sweet Spot (55%):** The 55% variance target is the top performer across nearly all metrics, achieving the highest Sharpe Ratio (0.95) and the highest Annualized Return (6.18%). This suggests that 55% explained variance captures enough systematic risk to cleanly isolate idiosyncratic.
- **Under-fitting (40%):** At lower variance levels (40%), the strategy suffers from higher volatility and a significantly larger Maximum Drawdown (-15.24%). In this regime, the residuals likely still contain significant unhedged systematic risk, leading to higher correlation between positions.
- **Over-fitting (65% $-$ 75%):** As we increase the variance target beyond 55%, performance begins to degrade. By explaining 75% of the variance, we are likely treating specific, mean-reverting idiosyncratic signals as factors. This reduces the profit opportunities, resulting in lower total returns (53.43%), though it does result in the lowest volatility (5.80%) as the portfolio becomes more market-neutral.

Looking at the cumulative returns chart, all variable strategies outperform the fixed 4-factor baseline in the long run, as the latter has not a statistical foundation. We can conclude that a dynamic approach to factor selection is superior to a fixed one, with the 55% variance target providing the optimal balance between systematic risk removal and the preservation of alpha-generating idiosyncratic signals.

7 Trading Time: Volume-Adjusted Returns

7.1 Methodology

Following Section 6 of the paper, we extend the baseline strategy by incorporating daily trading volume information into the signal generation process. The key intuition is that price dislocations occurring on high volume are more likely to reflect genuine fundamental repricing and should not be faded, whereas the same move on thin volume is more likely to represent a temporary anomaly susceptible to mean reversion.
To formalise this idea, we replace the standard calendar-time returns with *trading-time* returns, defined as:

$$\bar{R}_{i,t} = R_{i,t} \cdot \frac{\langle \delta V_i \rangle}{V_{i,t}} \tag{10}$$

where $\langle \delta V_i \rangle$ denotes the trailing 60-day average daily volume for asset i, and $V_{i,t}$ is the realised volume on day t. When daily volume is below its historical average, the adjustment factor exceeds unity and the return is amplified; when volume is elevated, the factor falls below unity and the return is dampened.

7.2 Results

Table 8 reports the performance statistics for the trading-time strategy alongside the calendar-time baseline, both evaluated without transaction costs and using a fixed $K = 4$ PCA factors.

Metric	Calendar Time	Trading Time
Total Return	49.23%	58.81%
Annualized Return	4.40%	5.23%
Annualized Volatility	6.06%	6.07%
Sharpe Ratio	0.73	0.86
Maximum Drawdown	-11.59%	-12.32%

Table 8: Performance Comparison: Calendar Time vs. Trading Time

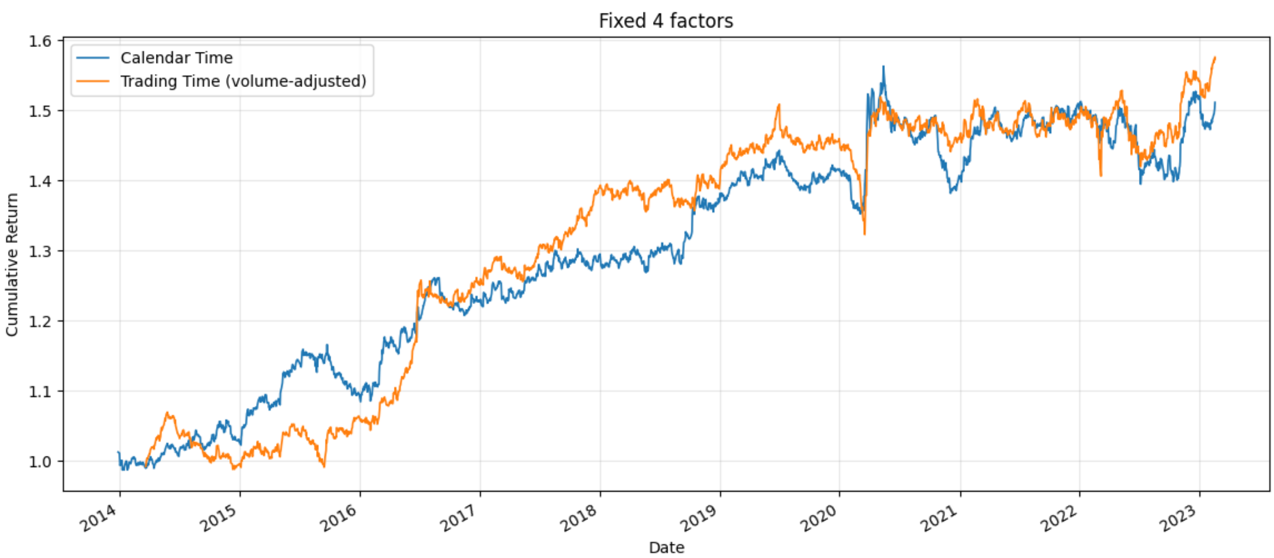

Figure 9: Cumulative returns of the calendar-time and trading-time strategies.

7.3 Discussion

The volume adjustment delivers a meaningful improvement in risk-adjusted performance. The Sharpe Ratio increases from 0.73 to 0.86, an improvement of approximately 18%, driven entirely by higher returns (+0.83%

annualised) at essentially identical volatility (6.06% vs 6.07%). This result confirms that volume information adds genuine signal content beyond what is captured by price alone.

The slight increase in maximum drawdown (-12.32% vs -11.59%) is a modest trade-off for the higher return. The drawdown remains well-contained relative to the Equal Weight benchmark (-39.93%), confirming that the market-neutral structure of the strategy is preserved under the volume adjustment.

Summarizing, the trading-time extension is a worthwhile enhancement to the PCA strategy on our dataset. The results are broadly consistent with the paper's finding that volume information is valuable in the mean-reversion context.